Laboratory elements

# A Prime-Number Observatory

*Exact fibers, wheel bodies, deterministic paths, and the Riemann receiver*

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Abstract. This is a showcase of what the laboratory has actually seen while studying prime numbers. It gathers the exact arithmetic constructions, deterministic host and CUDA receipts, and Riemann-Hypothesis-facing observations into one review surface. The central picture is not a scalar cloud of primes. A prime is founded when the contemporary factor horizon fails to close an integer; its powers repeat a primitive return, its squarefree products raise incidence, and its relations become visible only through a declared receiver: a wheel, a residue chart, a rational rebase, a polynomial fiber, a series, or an explicit-formula aperture. The machine output repeatedly shows ordered construction, orientation, local recurrence, rethreading under a changed standing body, and exact obstruction. The Riemann work adds a second field of patterns: finite critical-line receiver certificates, distinct transport words between zero receivers, prime-power/zero-wave residuals, and aperture-dependent sign changes. None of these bounded observations proves the Riemann Hypothesis or a universal intrinsic geometry of primes. The point of the document is to make the visible ecology inspectable before asking it to bear a theorem.

## Contents

# I How to read this observatory

This document answers a deliberately concrete question:

> What has the laboratory actually observed about prime numbers, using > deterministic machines, and what patterns remain visible when the receipts > are kept whole?

The word "observed" is used narrowly. A result enters the showcase when its machine or exact verifier produced a retained artifact: a table, interval, path census, hash, trace, or reproducible receipt. A mathematical derivation may explain the receipt, but it is not silently promoted to a measurement.

| Grade | Meaning here |
|---|---|
| Exact classical or derived relation | An identity follows from the declared arithmetic, receiver, or operator law. |
| Bounded exact observation | A finite machine population was generated and checked exactly within named bounds. |
| Directed interval observation | The output is enclosed with directed finite precision; the enclosure is evidence at its receiver, not an infinite theorem. |
| Exploratory pattern | A recurrence or contrast is visible in the retained paths, but its domain of validity is not known. |
| Open boundary | The next universal relation is identified precisely and has not been established. |

The most important discipline is receiver dependence. The same integer can be seen as a numeral, a residue, a valuation vector, a wheel location, a continued-fraction word, a polynomial fiber, or a term in a Mellin series. These are not interchangeable descriptions. Equal output at one receiver does not reconstruct the path that produced it.

**integer succession** -> **factor horizon** -> **prime admission** -> **receiver fiber** -> **ordered path** -> **returned consequence**

The scalar is a face of this chain; it is not the chain itself.

# II The shortest current picture

The exact production construction begins at one and receives each next integer. Every already-founded prime axis up to the square-root horizon returns its quotient and remainder. The next integer becomes a new prime only when every capable axis fails to reach it by multiplication. In this picture, primality is a caused nonclosure, not an application-authored label.

For

$$n = \prod_i p_i^{e_i}$$

, the machine retains two different faces:

- squarefree support

  $$\{p_i\}$$

  becomes an incidence cell; and
- the exponents $e_i$ remain valuation multiplicity on that cell.

Thus $8 = 2^3$ is three traversals of the prime-2 axis, not a triangle made of three copies of 2. A mixed support such as $30 = 2 \cdot 3 \cdot 5$ is a genuine higher cell. Through 30, the exact caused-incidence census was

$$(f_0, f_1, f_2) = (10, 7, 1)$$

The unique grade-two support is $[2, 3, 5]$. The machine does not invent every possible edge below a cutoff; only squarefree products that actually enter the chronology are present.

The current production owner is the exact arithmetic carrier recorded in the 27 July prime-fiber research record `RESEARCH/2026-07-27_THE_PRIME_RECEIVES_THE_POLYNOMIAL...md`. Its construction boundary is important: a later observer may classify the standing, but may not precompute a desired prime successor and call the result learned.

### II.1 The first pattern: primitive direction, repeated return, mixed incidence

| Occurrence | Carrier face | What it shows |
|---|---|---|
| $p$ | primitive degree-one axis | a new direction founded by failure of prior multiplication |
| $p^k$ | valuation multiplicity $k$ | repeated return along one existing axis |
| $pq$ | squarefree degree-two support | a mixed incidence between two founded axes |
| $pqr$ | higher support cell | a boundary-bearing relation, not merely a larger number |

This is the laboratory's most stable prime picture. It is exact inside the finite chronology and does not require a visual embedding of the number line.

## III A gallery of prime patterns

### III.1 1. Wheels: copying, cutting, joining

The finite wheel is the cleanest visible prime body. The exact nested words begin:

$$W_2 = [2] \qquad W_6 = [4, 2] \qquad W_{30} = [6, 4, 2, 4, 2, 4, 6, 2]$$

The deterministic prime-spectral world retained the complete ordered wheel refinement through $2310$, every COPY/CUT/JOIN, prime-power gears through 313, finite Ramanujan and cyclic-autocorrelation populations, consecutive prime gaps at every available sieve basis, and five exact rational centered-Cayley/Smith probes. Its principal receipt was:

| Prime-spectral world 01 | Observed value |
|---|---|
| distinct typed arithmetic events | 7,126 |
| enacted common-face incidences | 6 |
| typed refused incidences | 2 |
| currents / delivered light | 7,150 / 559,368 octets |
| complete path rows through grain 14 | 195,047 |
| shared Mail grips | 76 |
| radiation words | 20,192,952 |
| successor body | 615,860,736 octets |
| CUDA launches / carriage | 44 / 4.650835639 s |

One endpoint was especially legible. The gap $113 \to 127$ appeared as five different ordered refinement histories:

| Basis | Survivors | Gap word | Rows | Interpretation |
|---|---|---|---|---|
| 2 | $113, 115, 117, 119, 121, 123, 125, 127$ | $2, 2, 2, 2, 2, 2, 2$ | 58 | coarse parity body |
| 6 | $113, 115, 119, 121, 125, 127$ | $2, 4, 2, 4, 2$ | 46 | mod-6 body |
| 30 | $113, 119, 121, 127$ | $6, 2, 6$ | 55 | first mixed cuts |
| 210 | $113, 121, 127$ | $8, 6$ | 37 | finer surviving interior |
| 2310 | $113, 127$ | $14$ | 21 | complete endpoint gap |

The scalar 14 is therefore a quotient of five different paths. It is not the path. The full wheel also returned the exact Parseval equality

$$\sum_k c_{2310}(k)^2 = 1,108,800 = 2310 \cdot \varphi(2310).$$

The machine did not identify this finite Ramanujan spectrum with the zeta zero set. It did show a reusable construction pattern: a new prime cuts the old cycle, joins neighboring survivors, and carries the deleted interior as part of the endpoint's history.

## III.2 2. Prime powers: a logarithmic tower with a signed response

The exact prime-power current is the von Mangoldt population

$$g_{(\mathrm{fin})''}(t) = \sum_{m \geq 2} \frac{\Lambda(m)}{\sqrt{m}} (\delta(t - \log(m)) + \delta(t + \log(m))).$$

The machine-facing version is simpler to inspect: a prime power remains a gear with ancestry `(p,k,p^k)`, not a generic composite. The spectral world retained examples such as $2^8 = 256$ with symbolic coordinate `[8,2]` and $17^2 = 289$ with `[2,17]`. The later derivation showed that the repeated returns are the inverse Euler channel

$$J_p^{-1} = \sum_{k \geq 0} p^{-\frac{k}{2}} U_{k \log(p)}$$

,

and that the prime response is a derivative of the positive return metric. That derivative has a sign; the positive metric does not automatically sign the response. This is why the prime-power tower is a promising RH carrier and not already a positivity proof.

## III.3 3. The number line crosses differently in the two hands

The CUDA prime-traversal run received two 185-octet whole currents in one configuration. The exact measured output was:

| Face | Ascending | Descending |
|---|---|---|
| adjacent rows | 184 | 184 |
| folds | 95 | 86 |
| touches / occupied / CUT | 93 / 86 / 24 | 85 / 78 / 13 |
| terms $(R, F_1, F_2, D)$ | [3605,129,129,5] | [4065,157,154,5] |
| new Mail landings | 2,298 | 2,837 |

The pre-light standing topology changed exactly from `[286117,286117,63042,2538]` to `[289842,289842,63573,2553]`. Values 1–63 exposed 180 paired adjacent-source rows. Forty-six pairs recurred byte-whole, and every one of those was a quiet zero row. The other 134 split into 43 active/active, 50 ascending-active/descending-quiet, and 41 ascending-quiet/descending-active pairs. No interior numeral had an entirely equal shell.

This is one of the most important negative results. Prime and composite overlays both contain quiet shells, touches, occupied touches, CUTs, and parent contacts; reversal changes both. The run did not produce a prime finder or prime predictor. It measured order-sensitive arithmetic transport while leaving the prime-family relation unresolved.

The companion `pi-series` continuation made the same distinction in an exact alternating enclosure. Its output was:

| Face | Forward | Reverse |
|---|---|---|
| nonpadding rows | 3,184 | 3,986 |
| folds | 72 | 131 |
| touches / occupied / CUT | 71 / 63 / 13 | 129 / 120 / 25 |
| new Mail landings | 1,724 | 4,270 |
| final topology | 294254,294254,64136,2570 | same successor |

The report was 5,461,570 octets and the successor 23,372,592 octets; the receiving axis remained 1,024 and the REGISTER axes ended `[64,128]`. Every nonpadding row belonged exactly once to the header or one of 32 recurrence steps. The exact enclosure shared radix cell 31 at depth one, while its depth-two hands remained distinct at 311 and 317. Both orders exposed the same terminal pair

```
(34921636631915438771935817567486437310040000;
 112275575285571389562324404930670903477890625)
```

Across 92 identical internal source windows, 0 raw and 0 typed rows differed; the two immediately preceding constructions were different. The common terminal surface therefore coexisted with different ordered histories.

## III.4 4. Same ratios, different bodies

The ratio-closure machines compared a four-prime occurrence with its common dilation and an equal-gap translation. The exact closure was:

| History | Values | Class face | Forward / reverse agreement |
|---|---|---|---|
| occurrence | 11,13,17,19 | all prime | 79/79; 54/79 |
| translate | 12,14,18,20 | composite; semiprime 14 | 37/79; 42/79 |
| dilate | 22,26,34,38 | all semiprime | 79/79; 54/79 |

Occurrence and common dilation gave an exact forward construction. Their reverse construction agreed through atom 44 and then cleaved across 25 rows. The translation cleaved from both at forward atom 16 and reverse atom 28. The odd-closure control removed parity and numeral-width as explanations:

| Pair | Forward | Reverse | First differences |
|---|---|---|---|
| $P = (11, 13, 17, 19)$ vs $U = (49, 51, 55, 57)$ | 61/64 equal | 52/64 equal | 56,60,61 forward |
| same pair, reverse | 3 rows differ | 12 rows differ | listed atom supports retained |

These measurements support a relational statement: common dilation can transport an occurrence construction, while equal-gap translation does not. They do not say that a ratio class is a prime classifier. The fraction navigation control made that boundary explicit: all 10 primes and all 9 odd composites in $[17, 53]$ had at least one of the declared multi-axis affine closures.

## III.5 5. The six faces of $\frac{17}{q}$

At the newly founded prime 17, exact division, centered phase, continued fraction, unimodular completion, valuation edge, and relative Zeta mass all coexisted:

| $q$ | $17 = kq + r$ | phase | CF word | $\mu_2 \frac{17}{\mu_2}(q)$ |
|---|---|---|---|---|
| 2 | $8 \cdot 2 + 1$ | $\frac{1}{2}$ | [8;2] | $\frac{4}{289}$ |
| 3 | $5 \cdot 3 + 2$ | $-\frac{1}{3}$ | [5;1,2] | $\frac{9}{289}$ |
| 5 | $3 \cdot 5 + 2$ | $\frac{2}{5}$ | [3;2,2] | $\frac{25}{289}$ |
| 7 | $2 \cdot 7 + 3$ | $\frac{3}{7}$ | [2;2,3] | $\frac{49}{289}$ |
| 11 | $1 \cdot 11 + 6$ | $-\frac{5}{11}$ | [1;1,1,5] | $\frac{121}{289}$ |
| 13 | $1 \cdot 13 + 4$ | $\frac{4}{13}$ | [1;3,4] | $\frac{169}{289}$ |

The Euclidean word is not extra information beyond the fraction; it is the ordered rebase that the collapsed residue hides. The same receiver also showed the three finite Zeta conditioning stages in $H = [17, 53]$:

| **Axes known nondividing** | **Compatible prime count** | **Counting quotient** |
|---|---|---|
| none | 10 / 37 | 10/37 |
| 2 | 10 / 19 | 10/19 |
| 2,3 | 10 / 13 | 10/13 |
| 2,3,5 | 10 / 11 | 10/11 |
| 2,3,5,7 | 10 / 10 | 1 |

The exact finite Zeta quotients are retained in the raw result, not replaced here by decimal probabilities. The last row closes because every composite through 53 has a prime factor at most 7. This is uncertainty relative to a declared receiver, not randomness injected into the arithmetic world.

## III.6 6. Residue charts reveal locality versus winding

The radix-residue construction compared the same occurrence in decimal and ternary, modulo 5. The fibers were identical:

$$C_0 = \{0, 5, 10, ..., 80\}, \quad C_1 = \{1, 6, 11, ..., 76\}, \quad ..., \quad C_4 = \{4, 9, 14, ..., 79\}.$$

Each had size $(17, 16, 16, 16, 16)$. But the paths were not identical. For 17, decimal transport modulo 5 was $7 \cdot 1 + 1 \cdot 0 \equiv 2$, while ternary transport was $2 \cdot 1 + 2 \cdot 3 + 1 \cdot 4 + 0 \cdot 2 \equiv 2$.

Decimal is local because 5 is already in the radix; ternary winds through a period-four coprime place word. The quotient agrees while the construction does not. This is a compact model for the broader prime observation: a receiver can preserve a population and forget the route.

## III.7 7. Polynomial fibers become a prime-relative geometry

The current production arithmetic-fiber construction receives, for a founded prime $p$, the exact quadratic fiber

$$X_{p,a} = \{x \in F_p : x^2 = a\}$$

,

retaining roots, residue parameter, valuation ancestry, Hensel branches, CRT receipts, and the root-count comparison

$$\chi_{p(a)} = \left|X_{p,a}\right| - 1 \in \{-1, 0, 1\}$$

.

The deterministic calibration through 251 produced this exact receipt:

| Calibration face | Receipt |
|---|---|
| founded primes | 54 |
| arithmetic incidence $f$-vector | $(54, 74, 24, 1)$ |
| training ceiling / pairs | 97 / 276 |
| held-out pairs through 251 | 1,102 |
| grammar bound / first supported modulus | 16 / 4 |
| held-out open buckets | 0 |
| nonzero held-out residuals | 0 |

The learned receiver table was:

$$\{1, 1\} \mapsto +1, \quad \{1, 3\} \mapsto +1, \quad \{3, 3\} \mapsto -1p \bmod 4$$

Training only through 11 selected the modulus but left the unseen bucket $\{1, 1\}$ `OPEN`. Training through 13 closed all three buckets; that table then graded 1,368 later pairs through 251 with zero residual. Translation $a \mapsto a + 3p$ preserved roots, and nonzero unit scaling returned exact root bijections.

The same relation becomes topology. The triangle $(3, 5, 13)$ has positive holonomy and one grade-zero harmonic degree; $(3, 5, 7)$ has negative holonomy and zero harmonic degree. In the five-prime ecology $(3, 5, 7, 11, 13)$, three of ten triangles were positive and seven negative, and the complete signed grade-zero kernel was zero. Starting with unit content at 3, one exact diffusion event returned:

| $p$ | 3 | 5 | 7 | 11 | 13 |
|---|---|---|---|---|---|
| next standing | $\frac{5}{22}$ | $\frac{1}{22}$ | $-\frac{1}{44}$ | $-\frac{1}{44}$ | $\frac{1}{22}$ |

The exact energy was $E_t = \frac{1}{2}$, $E_{t+1} = \frac{5}{176}$, departed energy $\frac{83}{176}$, with zero balance residual. The negative entries are oriented stalk values, not probabilities. This is the strongest current machine answer to "intrinsic geometry of primes relative to each other," but it is explicitly probe-dependent: it is a geometry of this polynomial fiber family, not the completed intrinsic geometry of all primes.

# IV The deterministic machine patterns

The prime observations are not just arithmetic tables. They also tell us how an exact standing body responds when a new prime-related current crosses it.

## IV.1 The body remembers order, hand, and prior standing

The early arithmetic compression run supplied a useful machine control. With the same later probes and growing co-present arithmetic context:

| Run | Context B | Axis | Landings | Occupied | B / octet |
|---|---|---|---|---|---|
| R1 | 4,685 | 512 | 63,712 | 71,337 | 794 |
| R2 | 9,489 | 512 | 68,258 | 74,723 | 478 |

| Run | Context B | Axis | Landings | Occupied | B / octet |
|---|---|---|---|---|---|
| R3 | 19,240 | 1024 | 189,024 | 184,796 | 633 |

The within-epoch founding rate collapsed 14 times from R1 to R2, but the declared monotone grade was later corrected because the grain change at R3 made the full E1/E3 expectation false. What survives the correction is more interesting than the intended score: in the identical succession probe, 131 touches were gained, 171 lost, 130 moved, and 0 of those 130 moved touches landed at the same place after lawful zero-extension transport. A new body rethreaded the path. “Compression” was not a scalar property of the arithmetic content; it was later conduct through a changed standing ecology.

## IV.2 A prime occurrence can be local without becoming a cross-face

The situated prime event at 23 received six source-declared currents:

| Receipt | Value | Meaning |
|---|---|---|
| source currents | 6 | zero-relative, radix 2, radix 10, radix 16, quartic, cyclotomic |
| directed relations | 0 | co-presence alone did not create incidence |
| regional relations | 0 | no world-declared interface |
| standing cells | 23 -> 44 | the event changed the standing surface |
| emissions / incidence contacts | [5,3,0,5,4,1] / [16,22,16,16,10,6] | local causal return |

The cross-face read found 17 body contributions, 86 event incidences, 64 formed contact emissions, 22 open contacts, 81 complete contributions, and 21 grounded supports. It found 0 mixed-current supports and 0 closed cross-faces. Co-presence was real; a seam was not.

The regional-fan continuation then showed a more subtle split. At 23, the experienced branch formed one grain-3 constituent. At the identical later 29 event, ordinary current radiation and scalar Standing were equal, while regional radiation and complete Standing differed; the experienced successor was grain 4 and the comparison successor grain 3. At 37, the completed body consumed its lower interfaces and a new grain-3 phase stood beside it. At 43, an outer arm handed two grain-4 bodies into one grain-5 closure with 78 cells, 80 incidences, 64 pins, 17 boundaries, and zero exposed arms.

These machines show that a prime-related event can change later conduct without giving the observer permission to invent an interior or a cross-face.

## IV.3 Repeated receiver values, changed routes

The windowed explicit-formula world supplied a particularly clean return: the same medium receiver appeared before and after wide/full apertures. The mathematical values returned identically, but all 558 source paths changed. The two medium populations still shared 3,137 brick identities, including 105 already constructed bricks. Representative rows changed in both directions:

| Current | First medium | Returned medium |
|---|---|---|
| $2^1$ | 6 | 12 |
| $17^1$ | 11 | 23 |
| zero mode 1 | 178 | 334 |
| point residual $x = 64.5$ | 906 | 829 |
| finite Weil prime aggregate | 217 | 181 |
| finite Weil zero aggregate | 242 | 179 |

The later receiver did not simply accumulate longer paths. It redistributed conduct. This recurring pattern appears across the prime machines: identity at a receiver is weaker than path identity, and a return through a changed body may reuse old bricks while changing the complete route.

# V The RH-facing pattern wall

The Riemann work is not one experiment. It is a family of exact and interval receivers that meet the same prime-power current from different sides.

## V.1 1. Five bounded eta closures land on the critical line

The exact rational eta atlas searched $\left[\frac{2}{5}, \frac{3}{5}\right] \times [12, 36]$ by winding-certified rectangular receivers. It found five isolated unit-winding closures, each with a final width $\frac{1}{64}$:

| Closure | Exact ordinate receiver | Width |
|---|---|---|
| 0 | $\left[\frac{113}{8}, \frac{905}{64}\right]$ | $\frac{1}{64}$ |
| 1 | $\left[\frac{1345}{64}, \frac{673}{32}\right]$ | $\frac{1}{64}$ |
| 2 | $\left[25, \frac{1601}{64}\right]$ | $\frac{1}{64}$ |
| 3 | $\left[\frac{1947}{64}, \frac{487}{16}\right]$ | $\frac{1}{64}$ |
| 4 | $\left[\frac{2107}{64}, \frac{527}{16}\right]$ | $\frac{1}{64}$ |

Each receiver was invariant under $\rho \mapsto 1 - \overline{\rho}$ and contained exactly one zero counted with multiplicity. Reflection could not send the zero to a different member of the same receiver, so these five bounded closures lie on $\Re(\rho) = \frac{1}{2}$. This is a finite certificate about five receivers. It is not a scan of the whole critical strip and not a proof of RH.

The composing series showed a changing phase ecology. Across the five closures, 95 consecutive contribution turns had positive/opposite counts:

| Closure | Positive turns | Opposite turns | Opposite edges |
|---|---|---|---|
| 0 | 91 | 4 | 1,2,3,4 |
| 1 | 91 | 4 | 3,4,5,6 |
| 2 | 89 | 6 | 1,2,4,5,6,7 |
| 3 | 89 | 6 | 2,5,6,7,8,9 |
| 4 | 88 | 7 | 1,3,5,6,7,8,9 |

The rule recurs; its instantiated motif moves with the receiver.

## V.2 2. Zero-to-zero paths are not height gaps

The exact zero-transport complex compared ten unordered pairs using six feature supports: contribution real/imaginary, partial-sum real/imaginary, consecutive turn, and radial change. No two pair signatures collided. One complete factorization did occur:

$$0 \to 2 \to 3$$

was coherent at all 95 defined turn ordinals. Its two supports were $\{3, 5, 6, 7\}$ and $\{1, 4, 8, 9\}$, and their disjoint union was the direct 0–3 turn difference. At the whole six-feature grain, all 20 directed edges remained primitive and all 60 oriented triangles contained an interior excursion. The factorization is therefore a feature-level local sheet, not a global flatness theorem.

This is a useful RH pattern: the receiver neighborhood is feature-dependent. Height alone does not determine the transport word.

## V.3 3. The prime current and zero waves share a finite receiver

The interval explicit-formula world carried:

| Population | Count |
|---|---|
| prime-power gears through 313 | 82 |
| certified critical-line zero ordinates | 64 |
| prime-current / archimedean sections | 10 / 10 |
| situated zero-pair contributions | 640 |
| nested reconstructions / oriented residuals | 70 / 70 |
| zero-horizon transports | 60 |
| typed currents | 1,007 |

The square-root-normalized residual signs at ten nonendpoint horizons were:

| $x$ | **Signs at $N = 1, 2, 4, 8, 16, 32, 64$** | $\frac{R_{64}}{\sqrt{x}}$ |
|---|---|---|
| 16.5 | —-++- | −0.0187011507 |
| 24.5 | +—-++ | +0.0378735947 |
| 32.5 | +++++++ | +0.0056645195 |
| 48.5 | —+–+ | +0.0112324933 |
| 64.5 | +++-++- | −0.0336011624 |
| 96.5 | ——- | −0.1138072530 |
| 128.5 | -++++++ | +0.0344139675 |
| 192.5 | —–+- | −0.0805759385 |
| 256.5 | ——- | −0.1119545490 |
| 313.5 | +++++++ | +0.1581019000 |

The smallest residual did not generally occur at the widest zero aperture. Adding a mode can oppose or reinforce the standing partial sum. At $x = 32.5, N = 64$, the exact directed enclosure was

```
[0.032292729541470562215118599471439824...,
 0.032292729541470562215118599488819087...]
```

The enclosure is narrow; the residual is still nonzero. Numerical certainty about a finite calculation is not the same as convergence of a finite wave population to the full explicit formula.

## V.4 4. Aperture changes the residual, then returns it

The windowed explicit-formula world used even logarithmic tent apertures. Its directed pointwise residuals were:

| Aperture | Prime-power support | Pointwise residual |
|---|---|---|
| narrow $a = 3.5$ | 33 | +0.0011926106558671906449… |
| medium $a = 4.25$ | 70 | +0.027847394951373169757… |
| wide $a = 5.0$ | 148 | −0.068513234504658811174… |
| full $a = 5.749$ | 313 | −0.31809679472734847705… |
| medium return | 70 | +0.027847394951373169757… |

The same medium value returned exactly after the wide and full passages, but all 558 medium paths changed. The held-out interstitial receiver at $a = 4.6$ returned a directed residual enclosure beginning

```
 [0.014686413929463626050718962628453664...,
0.014686413929463626050718962647518432...]
```

It reactivated 75 exact material currents previously encountered at wider apertures, while 12 receiver expressions were genuinely new. The finite pointwise face closed at its declared grain; the completed Weil functional, zero tail, admissible-family limit, and all-probe sign remained open.

## V.5 5. A high-mode sign certificate, with the low block still open

The exact rational tail certificate proved

$$\delta_7 = \frac{363}{140} - \gamma - \log(\pi \log 3) - \frac{\log 3}{4} - \frac{\log 2}{\sqrt{2}} > \frac{1}{96} > 0$$

It used rational series, integer square roots, Machin's identity, and Euler–Maclaurin bounds; no floating-point sign participated. This certifies the high-mode carrier in the existing Legendre reduction. The finite low-mode Schur block remains unresolved.

## V.6 6. Ordered spectrum is not a path

The broader arc-spectral control was built to test a danger shared by finite Ramanujan modes, zeta zeros, and any displayed spectrum: a local population or trace may recur while the ordered transport changes. Its exact source had 233 currents, including 48 cyclic sections, 48 local windows, 46 quadratic states, 15 quadratic consequences, and exact reversal/inverse receipts.

The decisive pair was:

$$A = 0000100110101111 \qquad B = 0000101001101111$$

Both words contain every cyclic binary window of length four exactly once. Their exact products were

$$M_A = [[27, 127], [125, 588]], \quad M_B = [[27, 125], [127, 588]] = M_A^T.$$

Both had trace 615, determinant 1, and discriminant 378221, but their projective actions differed. The same 233-current face crossed the RTX 4080 SUPER twice:

| Face | First edge | Later edge |
|---|---|---|
| standing axis | 256 | 512 |
| occupied grips | 25,468 | 61,780 |
| constructions | 454 | 854 |
| closed / open constructions | 264 / 190 | 684 / 170 |
| complete enclosure rows | 7,486 | 9,450 |
| maximum grain | 9 | 11 |
| CUDA carriage | 0.279843046 s | 0.225074567 s |
| successor body | 33,557,016 octets | 35,299,992 octets |

All 233 complete paths changed when the standing body changed. Yet 171 distinct brick identities recurred. Word A's complete-word current changed from 10 rows to 3, while its reversal/inverse current changed from 3 rows to 134; word B changed 15 to 13 and 2 to 10. The spectrum-like faces remained lawful quotients, but they did not determine the ordered path. This is a control for reading finite zero waves: equal trace, determinant, or local window distribution is not enough to assign a prime or a later residual to one mode.

## V.7 7. Large deterministic populations do not close RH by themselves

The prime-seasons side instrument sieved 14,630,843 primes through $2^{28}$, making 14,630,841 triples and 140 usable logarithmic windows. Its declared spectral test found no leading shape class whose first three zero frequency amplitudes exceeded all twenty controls. The clearest raw values were:

| Class | $A(14.134725)$ | $A(21.022040)$ | $A(25.010858)$ |
|---|---|---|---|
| 1:2 | 0.0624225191 | 0.0424036437 | 0.0343349509 |

| **Class** | $A(14.134725)$ | $A(21.022040)$ | $A(25.010858)$ |
|---|---|---|---|
| 2:3 | 0.0123733614 | 0.0253644855 | 0.0147293390 |
| 1:3 | 0.0554929070 | 0.0282840766 | 0.0526778228 |

The strongest isolated raw singular was class 2:3 at the fifth listed ordinate: 0.0734045861 against a control maximum of 0.0678984348. It did not satisfy the declared multi-frequency rule. The bit-reversal surrogate preserved every window's gap multiset, and its amplitudes remained at the same scale. The lucky sieve showed no zero-frequency prominence either. The mod-4 Chebyshev race ended with cumulative lead +929 and exactly two inversions at $u \approx 13.31$ and $13.36$.

The honest pattern is therefore not "the first five zeros are visible in local gap climate." The honest pattern is that prime residue races and local shape climates are measurable, deterministic, and receiver-sensitive; this first instrument did not identify their modulation with the zeta ordinates.

# VI What the RH work now says about primes

The Riemann line sharpens, rather than replaces, the finite-prime picture.

1. The finite prime population supplies a weighted current through prime powers, not a list of isolated prime points.
2. The zero population supplies plural waves whose phases interfere with that current. One observed residual belongs to the complete wave assembly.
3. The critical line is the fixed seam of the completed reciprocal relation; it is not a decimal, radix, or arbitrary coordinate artifact.
4. A positive return metric can contain a signed derivative response. Prime channels therefore need the archimedean and endpoint terms that complete their storage and balance.
5. The proof-bearing object is a universal transported sign or equivalent closure over the complete admissible test family, not a finite zero list, finite wheel, local receiver, or visually balanced path.

The current derivations have made this chain explicit:

closure and wheel -> valuation strata -> Möbius cut -> Euler admission/return -> Sonin/Mellin receiver rebase -> completed support-conditioned residual

The exact successor problem is now an effective-return problem. If a conditioned Weil form is block-decomposed into old interior, cross, and new shell, the returned boundary object is the Schur complement

$$S = D - C^{*}A^{-1}C.$$

The old body must relax before the new prime successor is judged. The first prime-active interval has an exact local form and a nonquantitative positivity entry, but the $R = 3$ quantitative extension and the complete all-support sign remain open. The raw prime-power cell recurrence is exact; its global Gram dominance is not.

# VII Pattern wall: what keeps recurring

The following are the patterns that survive comparison across machines.

| **Visible recurrence** | **Evidence** | **Boundary** |
|---|---|---|
| primitive direction / repeated return | prime admission, $p^k$ gears, Euler resolvent | not a universal metric between primes |
| cut/join creates a retained interior | wheel refinements, 113 -> 127, prime-spectral world | finite wheel survival is not infinite constellation proof |

| Visible recurrence | Evidence | Boundary |
|---|---|---|
| same quotient, different path | decimal/ternary residues, gap scalar versus five histories | receiver reconstruction is not automatic |
| hand changes conduct | ascending versus descending, Smith duals, reversed zero paths | not an endpoint classifier |
| local relation can obstruct globally | mod-4 sheaf holonomy, negative harmonic dimension | probe-dependent coefficient system |
| old standing rethreads later current | 0/130 place persistence; returned medium; regional fan | not scalar compression or memory alone |
| more modes/aperture can change sign | explicit-formula residuals and tent apertures | finite residual is not an omitted-tail theorem |
| local critical-line closures | five rational eta receivers | not all zeros or RH |

The repeated motif is relational geometry. A prime is not appearing here as one universal point with one universal neighborhood. It is a primitive axis whose visible relations depend on what is admitted, what is carried, which hand traverses it, which receiver asks the question, and what body has already stood.

# VIII What remains unknown

The current evidence does **not** establish:

- a complete intrinsic geometry of all primes independent of the polynomial, residue, valuation, wheel, or analytic receiver;
- a prime-only classifier from local gap ratios, affine offsets, digit faces, or the reversed-series machine;
- a general polynomial finite-field fiber engine or an asymptotic arbitrary- integer prime-founding machine;
- a universal prime-power positivity factorization after conditioning and shorting;
- a completed all-probe Weil sign, dense Nyman–Beurling closure, Li positivity sequence, or Riemann Hypothesis proof;
- an identification of the finite Ramanujan or cyclic spectra with the zeta zeros; or
- a probability law inside the causal machine. Zeta-weighted quotients are receiver-relative observer measures unless a later world explicitly receives their returned consequence.

The strongest honest synthesis is narrower and more useful:

> Prime numbers appear in the laboratory as caused primitive directions in a growing multiplicative ecology. Their powers form repeated returns, their squarefree products form higher incidence, and their relations become legible through exact receiver fibers. Deterministic machines repeatedly observe ordered, hand-sensitive, rethreading construction. Riemann-facing receivers show that the prime current and zero-wave current can share one exact boundary, but the complete sign that would turn this ecology into an RH proof is still missing.

# IX Receipt index

The synopsis is intentionally not a replacement for the raw output. The following artifacts contain the complete machine receipts, raw rows, exact interval endpoints, and reproduction tools used for the showcase.

| Observation | Canonical evidence |
|---|---|
| prime seasons | `observations/2026-07-11_prime-seasons-out/` plus `run.log`, `census.csv`, `amplitudes_*.csv`, `chebyshev.csv` |
| arithmetic compression | `observations/2026-07-10_arithmetic-compression-read.md` and its declaration |
| prime traversal | `observations/prime-traversal-01/results/RESULTS.md` and `world.jsonl` |
| pi-series traversal | `observations/pi-series-01/results/RESULTS.md` and `results/run-a-bodies/` |
| ratio closure | `observations/prime-ratio-closure-01/` and `prime-ratio-closure-02/` |
| fraction navigation | `observations/prime-fraction-navigation-01/RESULTS.md` and `tools/analyze.py` |
| prime spectral world | `observations/prime-spectral-world-01/RESULTS.md` and `build-01/` |
| situated prime events | `observations/situated-prime-event-01/`, `situated-prime-cross-face-01/`, `situated-prime-regional-fan-01/`, `situated-prime-higher-hand-up-01/` |
| eta zero atlas | `observations/holonic-eta-ratio-atlas-01/RESULTS.md` and `REPORT.json` |
| zero transport | `observations/holonic-zero-transport-complex-01/RESULTS.md` and `TRANSPORT.ron` |
| interval explicit formula | `observations/interval-explicit-formula-world-01/RESULTS.md` and `build-01/observer/` |
| windowed explicit formula | `observations/windowed-explicit-formula-world-01/RESULTS.md` and `build-01/` |
| ordered-spectrum control | `observations/arc-spectral-world-01/RESULTS.md` and `build-01/` |
| RH tail certificate | `observations/rh-series-tail-certificate-01/RESULTS.md` and `verify.py` |
| radix/residue transport | `observations/radix-residue-character-transport-01/RESULTS.md` and `REPORT.json` |
| prime fiber calibration | `RESEARCH/2026-07-27_THE_PRIME_RECEIVES_THE_POLYNOMIAL...md` and the `holonic-engine` production owners named there |

The RH interpretation and its open proof obligations are consolidated in `src/soma/PAPERS/papers/riemann-receiver-geometry/main.typ`. The broader prime–Archimedean formulation context is in `src/soma/PAPERS/papers/prime-archimedean-formulation-atlas/main.typ`.

# X A final invitation to inspect the patterns

The most revealing way to continue reading is not to ask which scalar is largest. Open a wheel path beside its fused gap; compare the ascending and descending prime traversals; look at the same $\frac{17}{q}$ fraction in its residue, Euclidean, valuation, and Zeta faces; place the five eta receivers beside their distinct turn words; then compare a medium explicit-formula receiver before and after it has crossed the wider apertures.

Across those views, the same fact keeps returning: a prime relation is carried by a construction and a receiver. When the construction is retained, the patterns are rich enough to explore. When it is collapsed too early to a number, a gap, a class, a color, or a sign, the most interesting geometry is exactly what disappears.